

P 3.2-11 Determine the power received by each of the resistors in the circuit shown in Figure P 3.2-11.

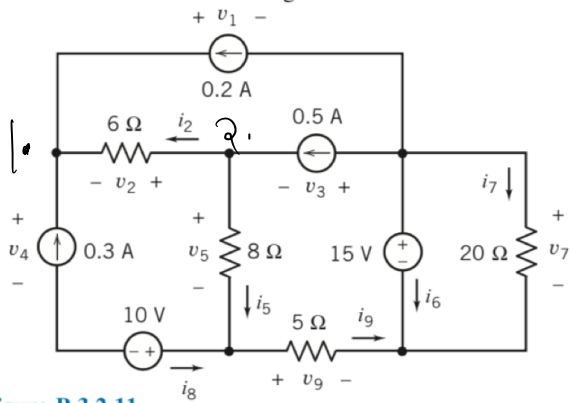


Figure P 3.2-11

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HW 3A
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1. KCL $0.3 \text{ A} + 0.2 \text{ A} + I_2 = 0 \rightarrow I_2 = -0.5 \text{ A}$
 $P_{6\Omega} = (-0.5)^2 (6) = \boxed{1.5 \text{ W}}$

2. KCL $-0.5 \text{ A} - 0.5 \text{ A} - I_5 = 0$
 $I_5 = -1 \text{ A}$ $P_{8\Omega} = (-1)^2 (8) = \boxed{8 \text{ W}}$

3. KCL $I_8 + I_5 + I_9 = 0$ $I_8 = 0.3$
 $I_9 = -0.7$

$P_{5\Omega} = (0.7)^2 (5) = \boxed{2.45 \text{ W}}$

4. $P_{20\Omega} = \frac{V^2}{R} = \frac{15^2}{20} = \boxed{11.25 \text{ W}}$

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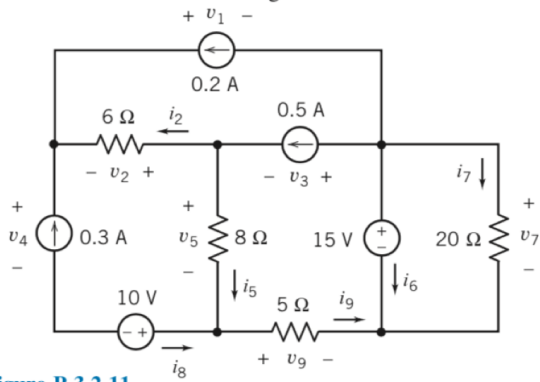


Figure P 3.2-11

$$I_1 = 0.75 \text{ A} + 0.25 \text{ A} = 1 \text{ A} \quad V_1 = 1 \text{ A} \cdot 20 \Omega = 20 \text{ V}$$

$$I_2 = 0.25 - 1 = -0.75 \text{ A} \quad V_2 = -0.75 \cdot 10 = -7.5 \text{ V}$$

$$V_3 = 15 \text{ V} \quad I_3 = \frac{15}{60} = 0.25 \text{ A}$$

$$15 \text{ V}$$

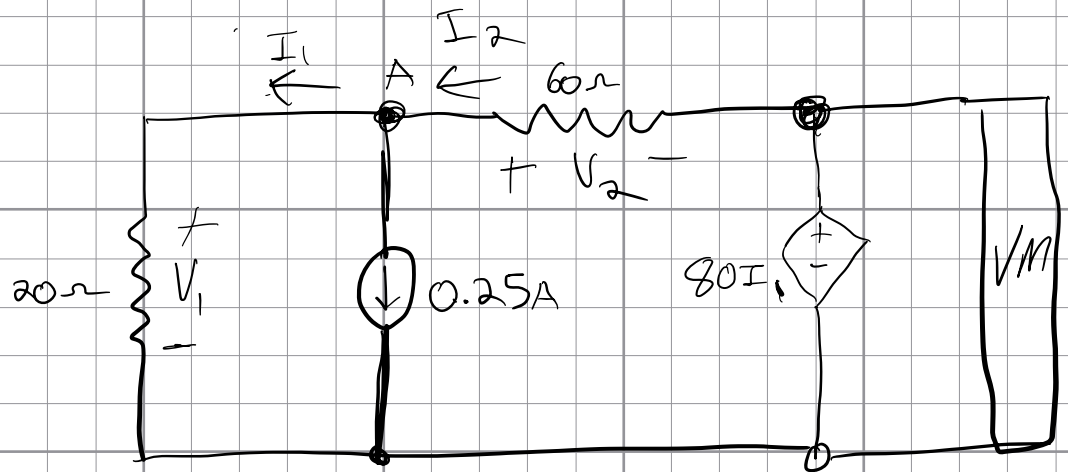
$$I_4 = 0.75 + 0.25 \text{ A} = 1 \text{ A}$$

$$0.75 \text{ A}$$



$$0.25 \text{ A}$$

$$V = 20 \text{ V} \quad V = 20 - 7.5 = 12.5 \text{ V} \rightarrow \text{parallel } 20 \Omega \text{ resistor}$$



Apply KCL at node A

$$I_2 = I_1 + 0.25$$

Find V_1 and V_2

$$V_1 = 20 I_1 \quad V_2 = 60(I_1 + 0.25)$$

Apply KVL

$$\begin{aligned} V_1 + V_2 + 80 I_1 &= 0 \\ 20 I_1 + 60 I_1 + 15 + 80 I_1 &= 0 \\ I_1 &= -0.09375 \text{ A} \end{aligned}$$

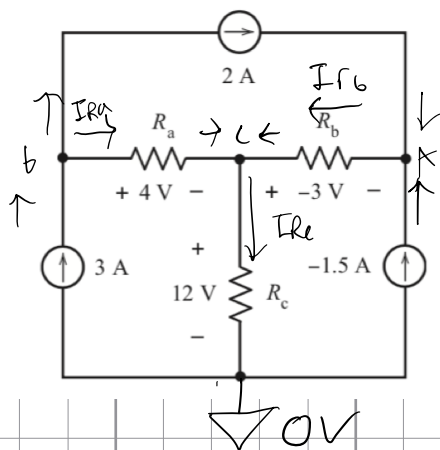
Solve for V_m

$$V_m = 80 I_1 = 80(-0.09375 \text{ A})$$

$$V_m = \boxed{7.5 \text{ V}}$$

P 3.2-27 $\oplus$ Consider the circuit shown in Figure P 3.2-27.

- Determine the values of the resistances.
- Determine the values of the power supplied by each current source.
- Determine the values of the power received by each resistor.



KCL @ Node A

(a) $I_{R_b} = 2 - 1.5 \text{ A} = 0.5 \text{ A}$

$$R_b = \left| \frac{-3 \text{ V}}{0.5 \text{ A}} \right| = 6 \Omega$$

KCL @ Node B

$$I_{R_a} = 3 - 2 = 1 \text{ A}$$

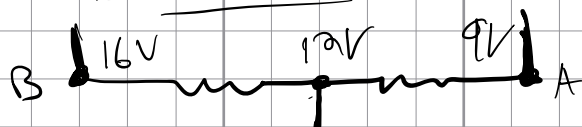
$$R_a = \left| \frac{4}{1} \right| = 4 \Omega$$

KCL @ Node C

$$I_{R_c} = 1 \text{ A} + 0.5 = 1.5 \text{ A}$$

$$R_c = \left| \frac{12}{1.5} \right| = 8 \Omega$$

(b) 2A Source



$$P = VI \quad \Delta V = 16 - 9 = 7 \text{ V}$$

$$P = 7 \times 2 = 14 \text{ W}$$

3A Source

$$\Delta V = 16 \text{ V} - 0 = 16 \text{ V}$$

$$P = 16 \times 3 = 48 \text{ W}$$

-1.5A Source

$$\Delta V = 9 - 0 = 9 \text{ V}$$

$$P = 9 \times 1.5 = 13.5 \text{ W}$$

(c) Ra

$$P = I_{R_a}^2 R_a = 0.5^2 \times 4 = 1.0 \text{ W}$$

Rb

$$P = I_{R_b}^2 R_b = 0.5^2 \times 6 = 1.5 \text{ W}$$

Rc

$$P = I_{R_c}^2 R_c = 1.5^2 \times 8 = 18 \text{ W}$$

P 3.2-28 $\oplus$ Consider the circuit shown in Figure P 3.2-28.

- Determine the value of the power supplied by each independent source.
- Determine the value of the power received by each resistor.
- Is power conserved?

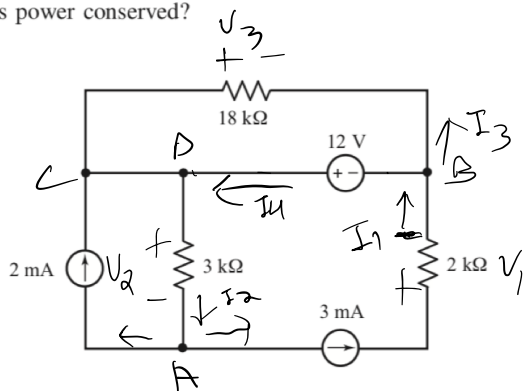


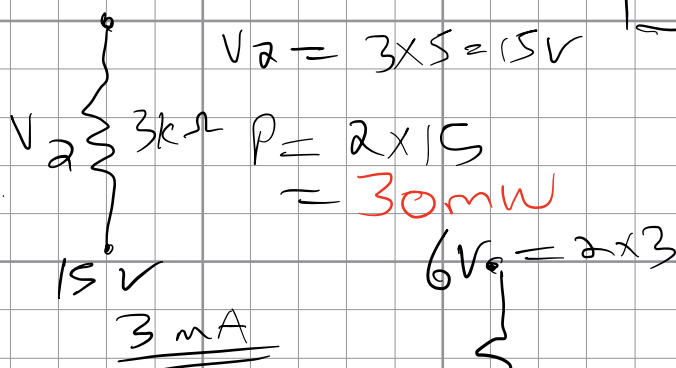
Figure P 3.2-28

(a) 12V Power supplied

$$P = IV \quad V = 12V \quad I_4 = 3 + \frac{2}{3} = \frac{11}{3} \text{ mA}$$

$$P = \left(\frac{11}{3} \times 10^{-3}\right) 12V = 44 \text{ mW}$$

2 mA



$$V_1 = 15 - 6 = 9$$

$$P = 3 \times 9 = 27 \text{ mW}$$

(c) Power-supply = power-received

$$44 + 30 + 27 = 8 + 75 + 18 \quad \checkmark$$

Power is conserved.

(b) 2 kΩ Power Received

$$P = I_1^2 R = (3 \times 10^{-3})^2 \times 2,000$$

$$P = 18 \text{ mW}$$

3 kΩ

KCL @ Node A

$$I_2 = 3 \text{ mA} + 2 \text{ mA} = 5 \text{ mA}$$

$$P = (5 \times 10^{-3})^2 \times 3,000$$

$$= 75 \text{ mW}$$

18 kΩ

$$V_3 = 12V$$

$$I_3 = 12 / 18k = \frac{2}{3} \text{ mA}$$

$$P = \left(\frac{2}{3} \text{ mA}\right)^2 \times 18,000$$

$$= 8 \text{ mW}$$

P 3.3-10 Determine the value of the voltage measured by the meter in Figure P 3.3-10.

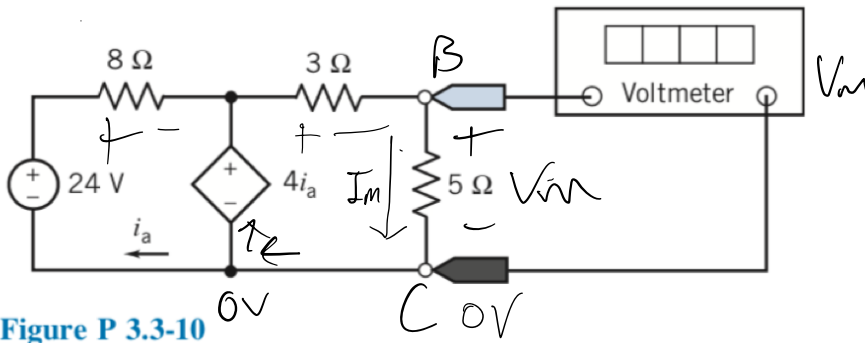


Figure P 3.3-10

Notes:

Current dependent Voltage Source

2 variables I_m , I_a

• Apply KCL + KVL

• circuit can be understood with mesh analysis.

KVL left mesh

$$-24 + 8i_a + 4i_a = 0 \quad i_a = 2A$$

KCL @ node A Right mesh.

$I_a + I_m \rightarrow$ Current flowing through 3Ω resistor.

KVL @ Right Loop

$$I_m(3\Omega) + I_m(5\Omega) - 8V = 0$$

$$I_m = 1A$$

$$V_m = 1 \times 5 = 5V$$

P 3.3-17 The input to the circuit shown in Figure P 3.3-17 is the voltage source voltage

$$v_s(t) = 12 \cos(377t) \text{ mV}$$

The output is the voltage $v_o(t)$. Determine $v_o(t)$.

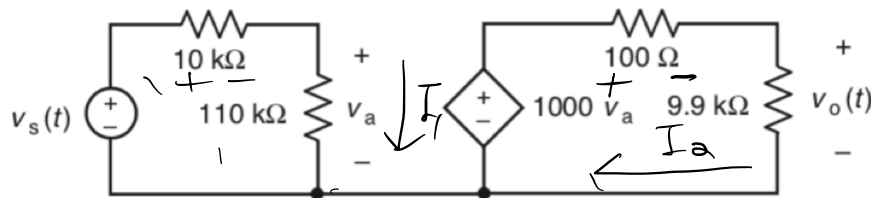


Figure P 3.3-17

$$V_{out} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_a = 12 \cos(377t) \times 10^{-3} \left(\frac{110k}{120k} \right)$$

$$V_a = 11 \cos(377t) \text{ mV}$$

$$V_o(t) = 1000 V_a \left(\frac{9900}{10,000} \right)$$

$$V_o(t) = 10.89 \cos(377t) \text{ V}$$

P 3.5-3 Determine the power received by each resistor in the circuit shown in Figure P 3.5-3.

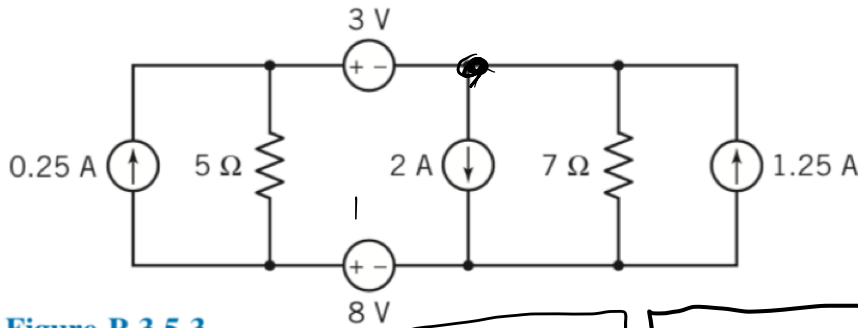
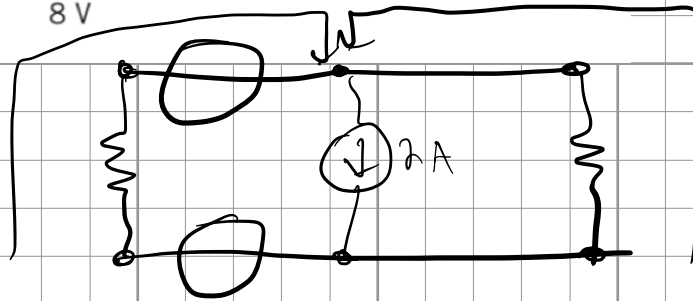


Figure P 3.5-3

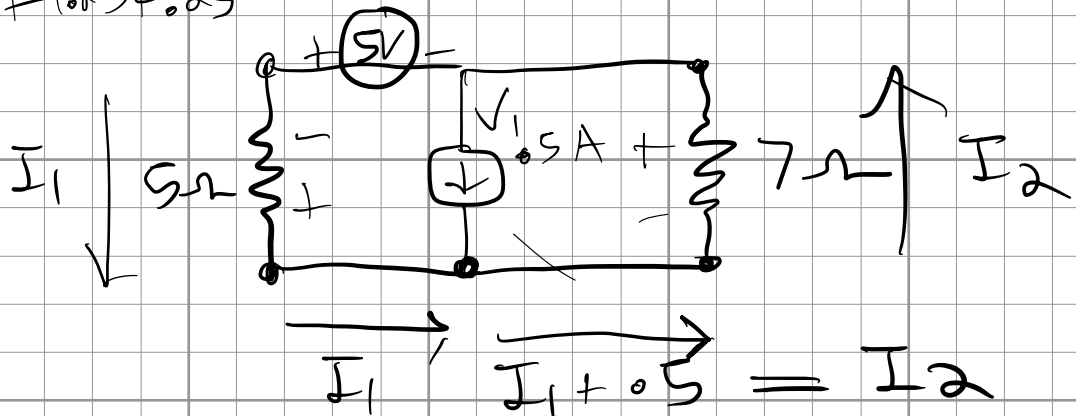


- This circuit is a current divider with 2 voltage sources in series.

- Combine sources and simplify circuit

$$I_T = -2A + 1.25 + 0.25$$

$$I_T = -0.5$$



KVL Left Mesh

$$+5 - 5I_1 = V_1$$

KVL Right Mesh

$$V_1 = 7I_2$$

$$P_{5\Omega} = 0.125^2 \times 5 = 7.8 \text{ mW}$$

$$P_{7\Omega} = 0.625^2 \times 7 = 2.73 \text{ W}$$

$$7I_2 + 5I_1 = 5$$

$$-I_2 + I_1 = -0.5$$

$$12I_1 = 1.5$$

$$I_1 = 0.125$$

$$I_2 = 0.625$$